



QI26- 20: Quantum Computing for Combinatorial Assignment Optimization

A Reproducible QUBQAOA Benchmark for Generalized Bipartite Assignment on NISQ Hardware

Verified progress · Real hardware evidence · Preliminary novelty assessment

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PHASE 1 · EXECUTABLE ARTIFACT



What the Project now demonstrates

Five claims supported by the executed artifact and the archived IBM Quantum job.

1 GMQAOA works ideally

Feasible probability = 1.0 and state preparation / full circuit fidelity = 1.0 on all three fixed instances.

2 Standard QAOA is functional

It recovers optimal scores in multiple settings; on the 42 capacity instance, feasible mass rises to 43.99% at p=3.

3 The pipeline reaches a QPU

Both p=1 circuits ran in one ibm_marrakesh Runtime job and sampled the exact optimum (score 3.25).

4 Rigor and traceability are built in

Same instances, score, and decoder; exact and Hungarian ground truth; exported CSV/JSON; fixed seeds; archived job ID; claims checklist and compliance matrix.

5 A preliminary originality result

Within the searched corpus, no prior publication was found that combines exact capacity assignment, standard vs. GMQAOA, controlled sampling, hardware execution, full distribution errors, and transpilation source analysis.

THE DEFENSIBLE MESSAGE

There is real, measurable progress: the algorithms function, the workflow is reproducible, and the experiment reaches present-day quantum hardware.



Generalized bipartite assignment as QUBO

One mathematical model, three controlled instances, one common scoring and validation layer.

DECISION VARIABLES

$x_{ij} \in \{0,1\}$ · agent $i \mapsto$ resource j

CONSTRAINTS

- Each agent is assigned exactly once.
- Resource use is either one-to-one or fixed by capacity vector C .
- All classical and quantum workflows use the same decoder.

IMPLEMENTED QUBO (exact capacity form)

$$Q(x) = - \sum_{ij} s_{ij} x_{ij} + \lambda \sum_i \left(\sum_j x_{ij} - 1 \right)^2 + \lambda_r \sum_j \left(\sum_i x_{ij} - C_j \right)^2$$

Penalty coefficients are generated programmatically; the QUBO-to-Ising mapping is numerically validated.

reference ~~33~~

9 qubits · 6 feasible

One-to-one assignment · optimum score 2.70

one to one ~~44~~

16 qubits · 24 feasible

Reproducible seed-2026 instance · optimum score 3.08

exact capacity ~~42~~

8 qubits · 6 feasible

Capacity vector $C=(2,2)$ · optimum score 3.25



Sevenstage reproducible research Pipeline

The notebook is an executable artifact, not a collection of disconnected code cells.



CONTROLLED COMPARISON

same instances + same score + same feasibility decoder + same backend protocol

Ground truth

Exact optimum and classical gaps

Algorithm quality

Feasible / optimal probabilities

Hardware quality

Counts, TVD, Hellinger, fidelity

Physical cost

Depth, instructions, native CZ



Strong classical reference points were established

The quantum results are interpreted against exact optimal or weak or ambiguous baselines.

3×3 oneto-one

2.70

GLOBAL OPTIMUM

Exact · Hungarian · Greedy · SA
all recover the optimum.

4×4 oneto-one

3.08

GLOBAL OPTIMUM

Exact and Hungarian are optimal.
Greedy = 2.80 · SA = 3.01.

4×2 exact capacity

3.25

GLOBAL OPTIMUM

Exact · Greedy · SA
all recover the optimum.

SCIENTIFIC ROLE OF THE BASELINES

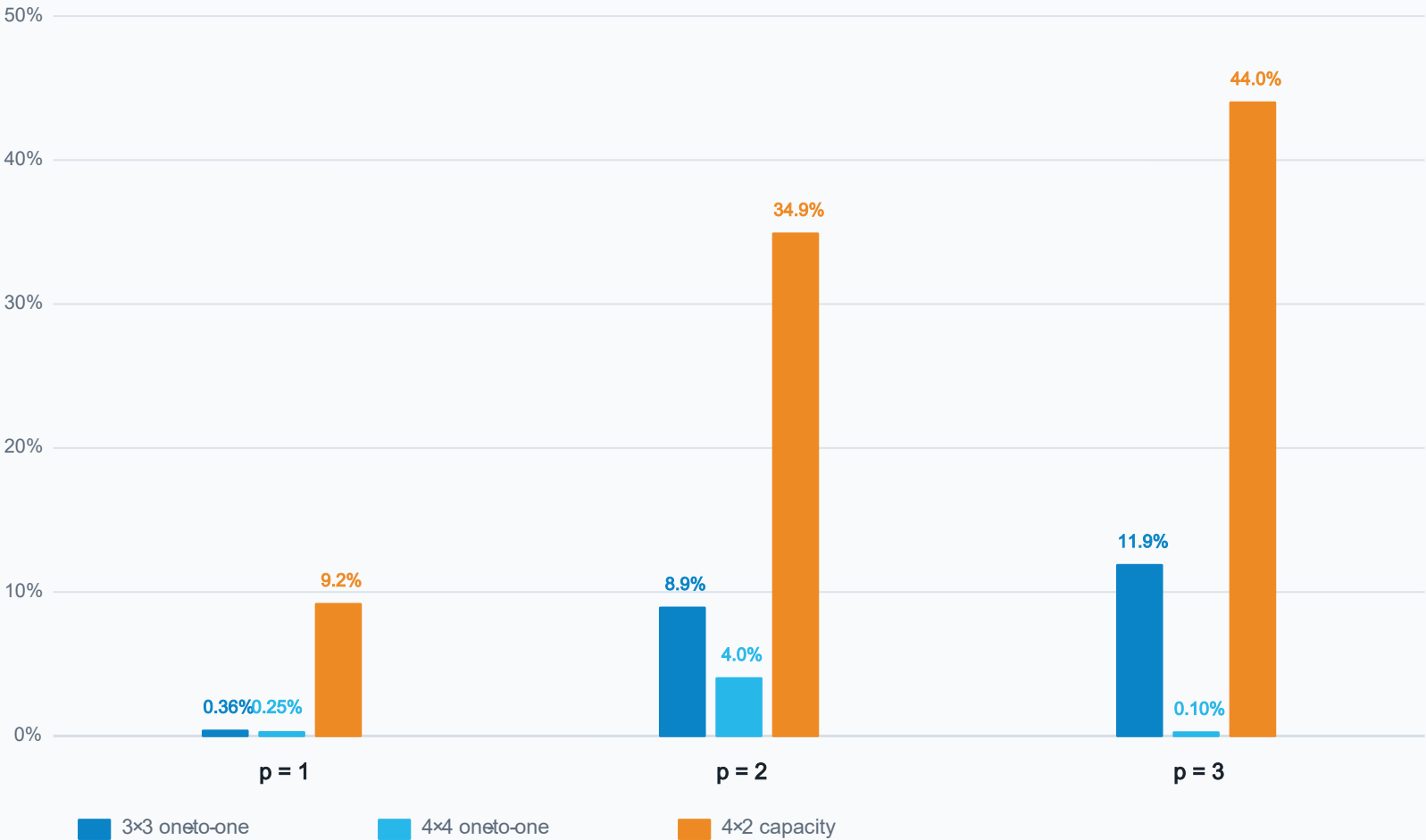
They establish the correct optimum, expose non trivial heuristic failures and make zero gap quantum samples interpretable.

Scope boundary: at these sizes, exact and Hungarian solvers are much faster.
This is a correctness and NISQ implementation benchmark, not a runtime competition.



Standard QAOA is functional in ideal simulation

Feasible probability by depth p . Labels show measured feasibility, while the right panel highlights the strongest tested settings.



3x3 · $p=3$

PF 11.86%

Popt 4.15% · best score 2.70

4x4 · $p=2$

Optimum 3.08

Only tested depth with zero gap

4x2 · $p=3$

PF 43.99%

Popt 6.11% · best score 3.25

Functional $\neq$ monotonic with depth p



GMQAOA works as an ideal feasible subspace Ansatz

The strongest verified result of Phase 1: feasibility and preparation fidelity are exactly preserved in ideal simulation.

reference 33

Popt 69.2%

PF = 100%
F(US) = 1.0
Circuit fidelity = 1.0
Best score = 2.70
Ideal depth = 1,062

one to one 44

Popt 10.7%

PF = 100%
F(US) = 1.0
Circuit fidelity = 1.0
Best score = 3.08
Ideal depth = 20,252

exact capacity 42

Popt 50.4%

PF = 100%
F(US) = 1.0
Circuit fidelity = 1.0
Best score = 3.25
Ideal depth = 395

WHAT IS PROVED

The implemented preparation + Grover mixer preserve the fixed feasible subspaces exactly.

SCOPE

Exact fixed-size constructions;
not a general scalable library.



The end-to-end pipeline reached real hardware

A controlled comparison: same backend, same job, same shots, same pass manager, same hardware options.

ibm_marrakesh · p=1 · 1,024 shots per circuit · optimization level 3 · dynamical decoupling · gate twirling

STANDARD QAOA

3.25

best feasible score
exact optimum · gap 0.00

Observed probability

PF 7.71%

Popt 1.56%

Physical resources

Depth 143

CZ 63

GMQAOA

3.25

best feasible score
exact optimum · gap 0.00

Observed probability

PF 11.91%

Popt 2.83%

Physical resources

Depth 1,357

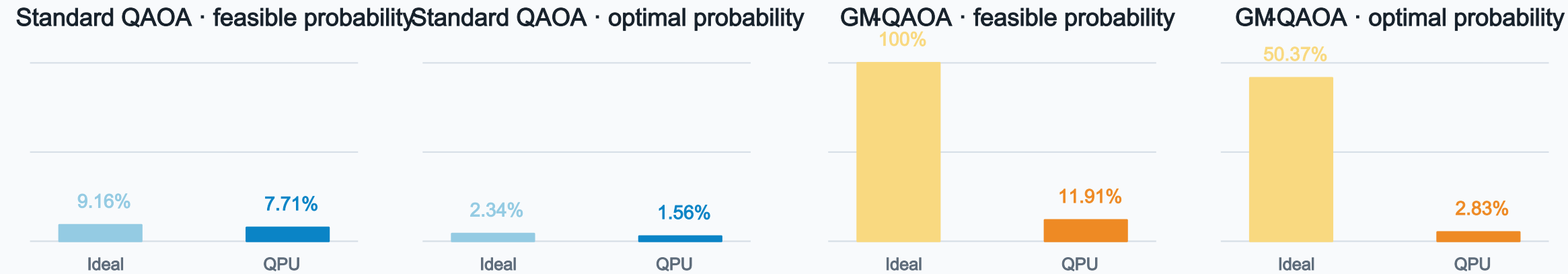
CZ 496

Both hardware distributions contained the exact optimum bits 10011001 · Runtime job d9vl8qfo3ppc73akfak0



Best-sample success is real, but it is not statistical reliability

The optimum appeared in both distributions, while the hardware concentration degrades especially for GMQAOA.



STANDARD QAOA

16 / 1,024 optimal-state counts

Its ideal optimum probability lies inside the physical Wilson interval.

GMQAOA

29 / 1,024 optimal-state counts

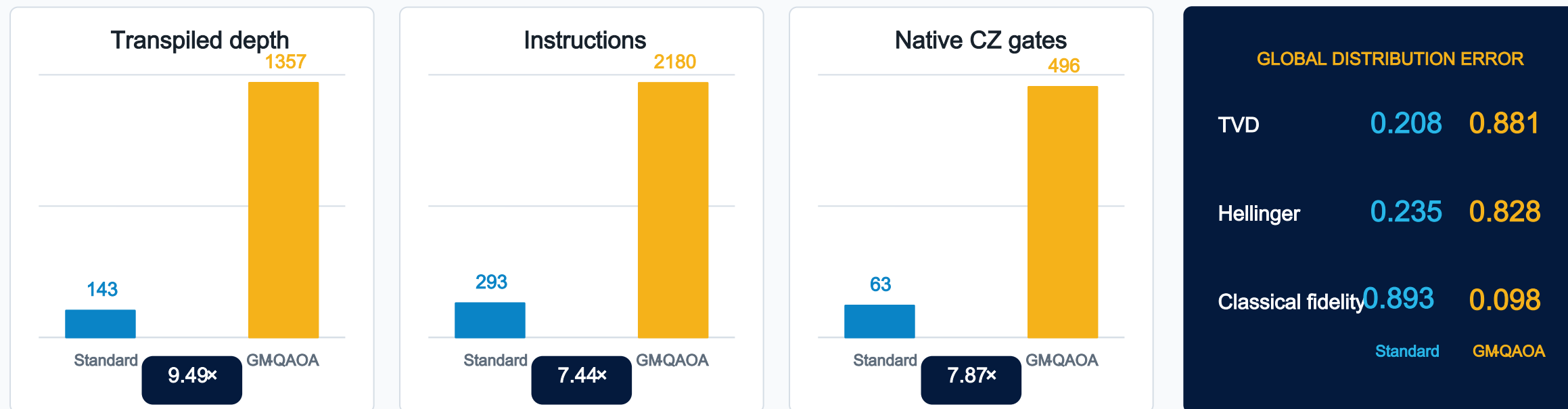
Its ideal 50.37% optimum probability is far outside the physical interval.

Correct interpretation: “the optimum was sampled” not “the optimum was found with high probability.”



The ideal GMQAOA advantage collapses under physical overhead

The data strongly support circuit length and synthesis/routing cost as major explanatory factors, proving depth alone is causal.



PRIMARY SCIENTIFIC INTERPRETATION

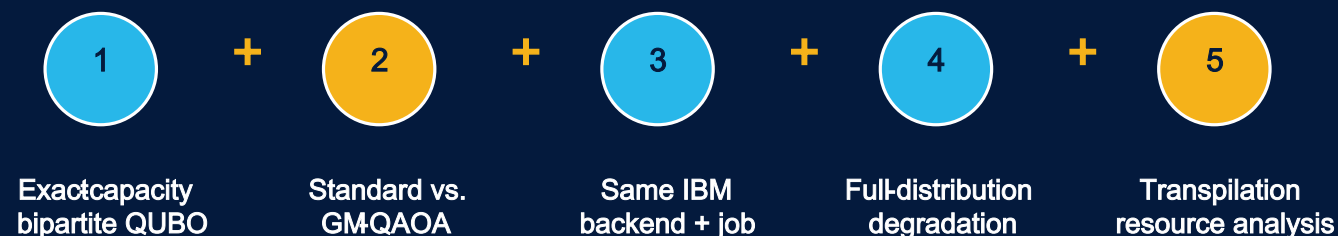
GMQAOA does not fail as an ideal ansatz. The specific fixed state preparation and sparse implementation are too costly to preserve the ideal advantage on this hardware execution.



A combinationlevel novelty resultstated cautiously

The individual ingredients have precedents; the searched corpus did not reveal the same controlled experimental combination.

THE EXPERIMENTAL INTERSECTION



“To the extent of the literature search conducted, this work presents an experimental combination not previously reported.”

Narrow claim: novelty lies in the controlled combination—not in inventing QAOA, GM-QAOA, QUBO assignment, IBM hardware execution, or the individual metrics.

CLOSEST PRECEDENTS

- Rhodes et al. (2025) —constrained vs. penalty mixer on IBM hardware
- Bärtschi & Eidenbenz (2020) —GM-QAOA method
- Stollenwerk et al. (2020) —hard-constrained assignment / resource estimates
- Golden et al. (2023) —Grover vs. transverse mixer scaling
- Hadfield et al. (2019) —alternating-operator framework

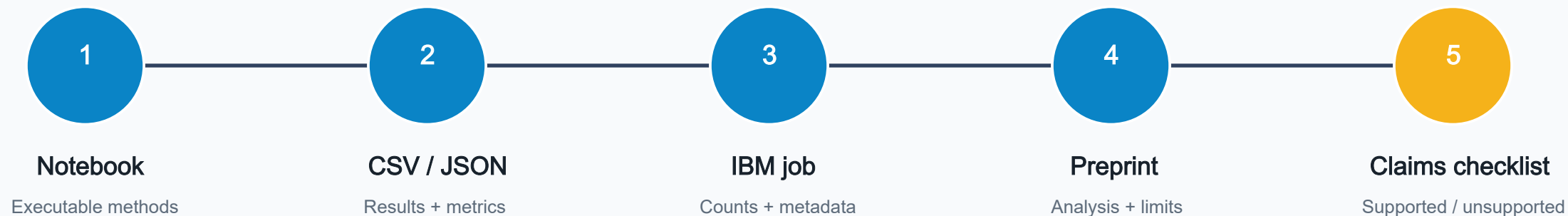
SEARCH CAVEAT

36 full-text articles were reviewed, but coverage was heterogeneous and not exhaustive of arXiv, IEEE Xplore, or Scopus. The conclusion remains provisional.



Evidence over hype: Every claim has an artifact boundary

This evidence-first framing aligns the presentation with QWorld's scientific integrity principles.



SUPPORTED BY THE ARTIFACT

- One-to-one and fixed exact capacity QUBOs are implemented and validated.
- Standard QAOA is compared at $p=1,2,3$ in ideal simulation.
- Fixed GQAOA circuits preserve the feasible subspace ideally.
- Both physical circuits sampled the exact optimum.
- The recorded GQAOA circuit has much higher depth and CZ count.

NOT SUPPORTED BY THE ARTIFACT

- Quantum advantage or practical runtime superiority.
- High-probability optimum recovery on hardware.
- QPU preservation of the GQAOA feasible subspace.
- A general scalable state preparation construction.
- Universal overhead ratios across devices or instances.



Phase1 Conclusions and the next research decisions

A balanced summary: verified achievements first, then the experiments required to generalize them.

1 GMQAOA works ideally

Exact fixed instance preparations preserve feasibility and produce much higher $p=1$ optimum probabilities.

2 Standard QAOA works

It recovers optimal scores in multiple settings; depth effects are instance and metric dependent.

3 The hardware pipeline works

Both matched $p=1$ circuits executed on ibm_marrakesh and sampled the optimum.

4 The physical tradeoff is measurable

GMQAOA's state preparation/mixer cost destroys most of its ideal distribution advantage.

5 The combination appears underreported

No exact methodological match was found in the searched corpus; this remains a cautious, provisional claim.

NEXT RESEARCH DECISIONS

01 Repeat matched jobs across time and multiple backends.

02 Replace generic fixed preparation with structured scalable circuits.

03 Run mitigation and ablation controls for DD, twirling, readout, and postselection.

04 Construct controlled size/capacity families and report total resource cost.

NO QUANTUM ADVANTAGE CLAIM

The contribution is a reproducible formulation, a controlled hardware experiment, and evidence based error/resource analysis.



THE TAKEAWAY

The feasible subspace construction works.
Standard QAOA works.
The complete pipeline reaches real hardware.
The evidence is reproducible and bounded.

Within the searched corpus, the controlled experimental combination was not previously reported.

No quantum advantage is claimed.

Phase 1 establishes a transparent foundation for the next experimental campaign.

<https://github.com/qubitmx/qi2620-quantumassignment-optimization>

PHASE 1

COMPLETE

QI2620
QUBO · QAOA · QAOA
IBM QUANTUM

THANK YOU!